

Sean Yépez · ME344 Summer 2026 · Final Project, Option 2

github.com/seanyeppez/me344ricechem

Problem: grading consumes the time meant for teaching

I love teaching, but of every 8 paid hours, 7 go to grading. I want to invert that ratio. Treemarks is my AI grading system for STEM classes: every decision checked against a rubric tied to the underlying logic, physics, and math.

AI checks every rubric decision → the TA reviews uncertain cases → students receive feedback sooner

RICECHEM · THE SUPERVISED BENCHMARK

RiceChem is the benchmark recent papers use: a chemistry exam graded against instructor rubrics. 1,264 responses · 4 questions · 27 rubric items → 8,392 TA-labeled TRUE/FALSE decisions. One decision = one answer + one criterion → the TA's label.

- [Sonkar et al. \(2024\)](#) · introduces RiceChem and rubric-based grading of long-form answers
- [Rao & Callison-Burch \(2026\)](#) · Autorubric, rubric-based LLM evaluation
- [Ferrer et al. \(2026\)](#) · when to trust LLM graders

RESEARCH QUESTIONS

RQ1 · Quality: can tuning make a local model accurate enough for rubric grading?

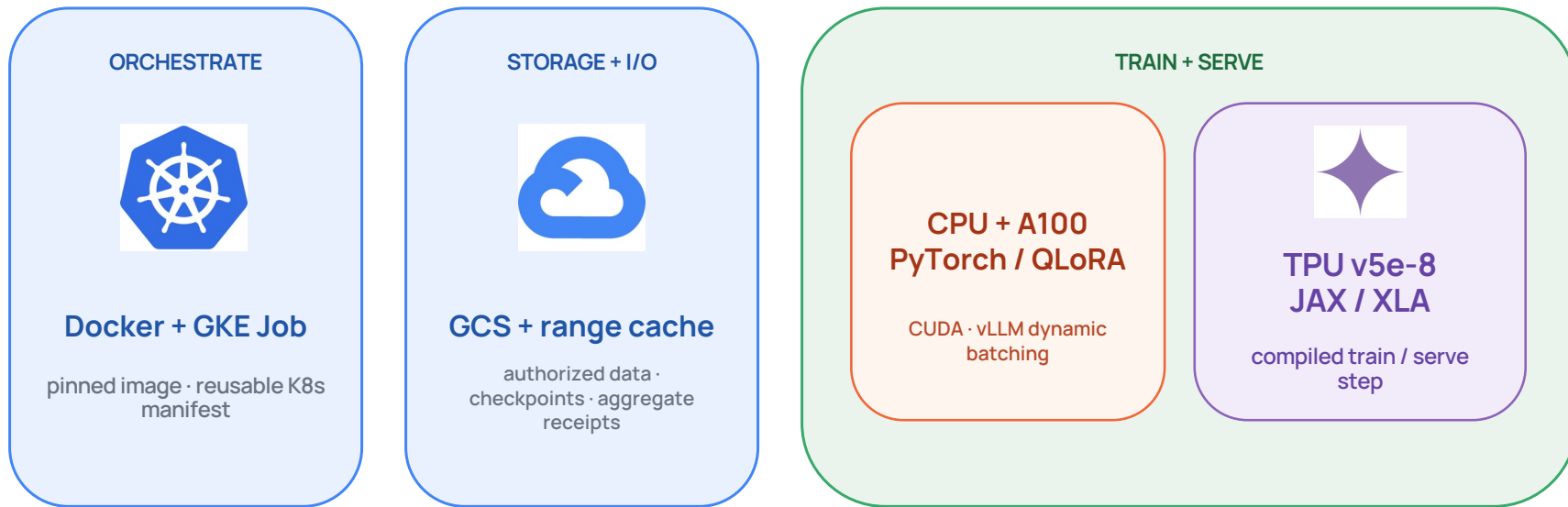
RQ2 · Systems: how do CPU, GPU, and TPU change latency, throughput, and utilization?

RQ3 · Operations: which configuration meets the quality bar in under 24 hours at lowest cost?

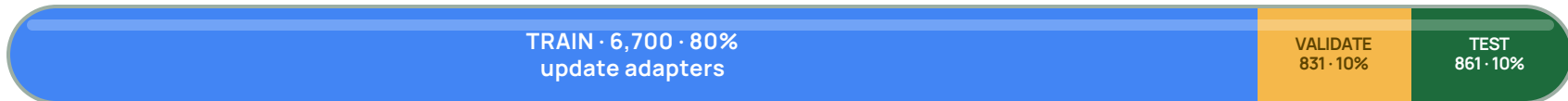


Sean Yépez · Dr. Mourad Bouache ·
Google

Containerize and run a supervised experiment on three compute lanes



EXPERIMENT DESIGN · 8,392 TA-LABELED DECISIONS



Validation picks the checkpoint. The test set is opened once, at the end.

Measurements: instrument the workload and disclose what is missing

CONTROLLED WORKLOAD

861 held-out rubric decisions

same manifest · prompt · parser · output cap · checkpoint

concurrency 1 + 24

EXECUTION LANES

RUNTIME

CAPTURED SIGNALS

CPU

16 vCPU baseline

Transformers

host memory · process telemetry

CPU utilization

peak RSS

p50 / p95

decisions / s

GPU

one A100 40 GB

PyTorch · CUDA · vLLM

NVML · batching · endpoint timing

GPU utilization

peak VRAM

load time

tokens + decisions / s

TPU

v5e-8 slice

JAX / XLA · vLLM-TPU

XProf · compile cache · GCS-FUSE

TPU utilization

peak HBM

compile time

cache + I/O

COMPARISON GATE

Compare hardware only when checkpoint, prompt, parser, output cap, batching, and concurrency match.

WORKLOAD

861 decisions · same
prompt/parser/output cap · c=1 and
c=24

COMPUTE

wall time · p50/p95 · decisions/s ·
wall/step time as cycle proxy

MEMORY

CPU mCPU + RSS · A100 NVML + VRAM ·
TPU host memory only

INPUT / I/O

GCS range cache fixed startup; load
and XLA compile time not isolated

Accelerators cut single-stream grading time 30–41×

THROUGHPUT SPEEDUP · CONCURRENCY 1 · CPU = 1×

16-vCPU



0.28/s · 50.9 min · 95.1% CPU · 18.0 GB RSS

A100 40 GB



8.54/s · 100.8 s · 19.5% GPU · 35.6 GB VRAM

TPU v5e-8



11.39/s · 75.6 s · device util/HBM unavailable · 54 GB host

CONCURRENT SERVING · c = 24

A100 123.9/s

\$0.007 per 861 decisions

TPU 98.4/s

\$0.023 per 861 decisions

These are one-endpoint measurements. Add GKE runners/endpoints as queue depth grows; cluster throughput is bounded by quota and coordination overhead.

QUALITY GATE · ~72% AGREEMENT

Gemma 4B failed the practical grading bar.

TA re-review overhead would erase the throughput benefit.

CONCLUSION

What this project changed for Treemarks grading

From prototype to cloud. A local prototype using local LLMs and frontier inference is now a hosted cloud solution running controlled ML experiments.

The feedback loop closes. Grades typically take a week or more to come back. A tuned 27B endpoint grades a full class set in about 51 minutes, so students could plausibly wake up to feedback the next morning.

The model lesson. Fine-tuning lifted the 27B's agreement 12 points, to parity with frontier models, at about 1/9 the serving cost, stable across five seeds. Gemma is a real candidate for grading going forward; the small 4B stays below the quality bar.

The operating rule. Meet the grading-quality bar first, then pick the smallest system that returns work within 24 hours, then minimize cost.

What limited us: unbatched 27B serving, accelerator memory, and storage startup pressure, each measured and mitigated in the report.

NEXT Batch the 27B endpoint · profile the 27B on TPU · test transfer to unseen questions · compare tuning economics with frontier inference.